

SCD Lab Business Case Charter: House Construction Progress Tracker

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Complex Computing Problem (CCP) Characteristics and Justification

1. Extent of Stakeholder Involvement

The system invoves three main stakeholders, and each has different responsibilities and requirements:

- **Homeowner:** Views the construction progress, completed and pending tasks, reported issues, and construction photos.
- **Supervisor:** Creates and updates construction tasks, changes progress percntages, reports construction issues, and uploads progress photos.
- **Admin:** Manages user accounts and basic project information.

Each stakeholder requires a different level of access. For example the homeowner should be able to view the progress but should not be allowed to change the supervisor's updates. The supervisor needs permission to updute construction information, while the admin has management permissions.

Therefore, the system needs to handle different stakeholder requirements and provide appropriate access and functions for each role.

2. Interdependence

The system contains several connected components that depend on each other, including:

- User login and authentication

- Role management
- Projects
- Construction phases
- Tasks
- Progress updates
- Issues
- Progress photos
- Dashboards
- Database

These components cannot work completely independently.

For example a project contains different construction phases, and each phase contains multiple tasks. Each task has its own progress and status. When the supervisor changes a task's progress from 20% to 30%, the new value is saved in the database, and the homeowner's dashboard retrieves and displays the updated progress.

Therefore, changes in one part of the system can affect other parts, so the components and database relationships need to be properly designed and integrated.

Business Case Charter Document:

House Construction Progress Tracker

1. Background

Building a house is a long process which includes many different types of work, such as foundation, brickwork, plumbing, electrical work, flooring, painting, etc. It is important to keep track of all these activities and their progress.

The House Construction Progress Tracker is a Java-based system that will help homeowners and supervisors keep track of the construction of a house. The homeowner will be able to see the progress of the construction, while the supervisor will be able to update the progress of different tasks. The information will be stored in a database so that the latest updates can be viewed by the homeowner.

2. Problem Statement

During house construction, it can be difficult for the homeowner to know the exact progress of the work. Usually, they depend on phone calls, WhatsApp messages, or verbal updates from the supervisor.

Some common problems are:

- It is difficult to track all construction phases.
- The homeowner may not know which work is completed or still pending.
- Delayed work may not be properly recorded.
- Construction issues can be difficult to keep track of.
- Previous progress information may be difficult to find.
- There is no single place where all project information is available.

The proposed system will solve these problems by providing one system for managing and viewing construction progress.

3. Business Objectives and Preliminary Requirements (SRS)

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The purpose of this system is to provide an easy way to track the progress of a house construction project. The system will allow the homeowner to view the progress of the construction, while the supervisor will be able to update the progress of different construction tasks.

All important information will be stored in a database so that updated information can be viewed by the users.

1.2 Scope

The House Construction Progress Tracker will be developed as a Java-based application connected to a database.

The system will have three main users:

- Homeowner
- Supervisor
- Admin

The system will allow users to manage and view construction projects, phases, tasks, progress, issues, photos, and basic expenses.

2. Overall Description

2.1 Product Perspective

The system will be a Java application connected with a database. After login, the system will identify the user's role and open the relevant interface.

The basic flow will be:

Login → User Role → Relevant Dashboard

For example:

Homeowner → Homeowner Dashboard

Supervisor → Supervisor Dashboard

Admin → Admin Dashboard

2.2 User Classes

Homeowner: Mainly views the construction progress and project information.

Supervisor: Updates construction work, progress, issues, photos, and other project information.

Admin: Manages users, projects, and basic system information.

3. Functional Requirements

Functional requirements describe what the system will do.

ID	Functional Requirement
FR-01	The system shall allow users to log in using their username and password.
FR-02	The system shall open a different dashboard according to the user's role, such as Homeowner, Supervisor, or Admin.
FR-03	The admin shall be able to add, update, and remove user accounts.
FR-04	The system shall allow construction work to be divided into different phases, such as Foundation, Structure, Electrical, Plumbing, and Finishing.
FR-05	The supervisor shall be able to add and update construction tasks under each phase.
FR-06	The supervisor shall be able to update the progress percentage and status of a construction task.

FR-07	The system shall save task and progress information in the database and display the latest information to the homeowner.
FR-08	The supervisor shall be able to add construction issues and update their status, such as Open, In Progress, or Resolved.
FR-09	The supervisor shall be able to add construction progress photos, and the homeowner shall be able to view them.
FR-10	The system shall display overall project progress, task status, and current construction information on the dashboard.

4. Non-Functional Requirements

Non-functional requirements describe how the system should work.

ID	Non-Functional Requirement
NFR-01	The system should respond quickly to normal actions such as login, viewing data, and updating tasks.(**Performance**)
NFR-02	Only registered users should be able to access the system.(**Security**)
NFR-03	Users should only be able to access the features allowed for their role.(**Security**)
NFR-04	The interface should be simple and easy to understand.(**Usability**)
NFR-05	The system should save information correctly and should not lose data during normal use.(**Reliability**)
NFR-06	The Java code should be properly organized so that changes and new features can be added easily.(**Maintainability**)
NFR-07	The system should store project information in a structured database with proper relationships
NFR-08	The system should validate important information before saving it.
NFR-09	The system should be available whenever the application and database are running.

NFR-10	The system should run on a computer with the required Java environment and database software.

5. Hardware and Software Requirements

Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM
- Sufficient storage space

Software Requirements

- Java
- NetBeans IDE
- SQL Server
- JDBC Driver

6. System Constraints

- The system will mainly focus on tracking house construction progress.
- Advanced features such as AI, live GPS tracking, online payments, and mobile applications are not part of the initial system.
- The system will depend on the database for storing and retrieving project information.

7. Future Enhancements

In the future, the system can be improved by adding:

- Mobile application
- Email or SMS notifications
- Cloud storage for photos
- Multiple construction projects
- More detailed financial management

- Online communication between homeowner and supervisor
- Advanced reports and charts
- Automatic delay alerts

4. Current Situation and Opportunity Statement

Current Problems

Currently, construction progress is often communicated through:

- Phone calls
- WhatsApp messages
- Verbal communication
- Paper records
- Excel files

This can make the information difficult to organize and track.

Proposed Solution

The proposed system will keep all construction information in one place.

For example, if a supervisor changes the progress of brickwork from **20% to 30%**, the new progress will be saved in the database. When the homeowner checks their dashboard, they will see the updated **30% progress**.

This will make the construction progress easier to monitor and keep updated.

5. Options and Recommendations

Option 1: Manual Tracking

The construction can continue to be tracked through calls, WhatsApp, paper records, or Excel.

Problem: It can be difficult to properly organize and maintain all the information.

Option 2: Use an Existing Project Management System if exists

An already available project management system can be used.

Problem: Such systems may not have features specifically designed for house construction.

Option 3: Develop a House Construction Progress Tracker

A separate system can be developed according to the needs of house construction.

Recommended Solution:
We will develop the House Construction Progress Tracker in Java with a database. It will have separate interfaces for different users and features for tracking construction phases, tasks, progress, issues, photos, and expenses.

6. Schedule Estimate

The project is expected to be completed within 4 months.

Work	Estimated Time
Requirements and SRS	**1–2 week**
System and Database Design	**1–2 week**
Interface Design	**3–4 week**
Java Development	**5-6 week**
Database Integration	**7-8 week**
Additional Features	**9-10 week**
Testing and Bug Fixing	**11–12 week**
Final Documentation and Deployment	**13-14 week**

Total estimated time: 4 months

7. Potential Risks and Mitigation Plan

Risk	Mitigation
The project becomes too large	Focus on the main features first and add extra features later.
Lack of experience in Java	Build the system step-by-step and start with simple features.
Database connection problems	Test the database connection at an early stage.
Data loss	Keep regular backups of the database.
Problems with storing images	Store images in a separate folder and save their file path in the database.
Project gets delayed	Divide the project into small tasks and complete them according to the schedule.
Bugs before final submission	Test each feature during development instead of testing everything at the end.

Note: Requirements and designs will change as we progress further in the project